

Experiment No.7

Demonstration of SSL/TLS (TLS 1.3) Protocol Using Wireshark in Kali Linux

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1.Client Hello

[illegible]

2. Cipher Suites

The image shows a Wireshark packet capture of a TLS handshake. The 'Packet Details' pane on the right highlights the 'Cipher Suites' field of a 'Client Hello' packet (packet 100). The list of cipher suites includes:

- TLS_AES_128_GCM_SHA256 (0xc02b)
- TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc030)
- TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (0xc032)
- TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc031)
- TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (0xc033)
- TLS_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)
- TLS_RSA_WITH_AES_256_GCM_SHA384 (0xc034)
- TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (0xc02c)
- TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA (0xc02d)
- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (0xc02e)
- TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA (0xc02f)
- TLS_RSA_WITH_AES_128_CBC_SHA (0xc02c)
- TLS_RSA_WITH_AES_256_CBC_SHA (0xc02d)

The 'Packet Bytes' pane on the right shows the raw data of the cipher suites field, which is a list of 13 cipher suite identifiers (0xc02b, 0xc030, 0xc032, 0xc031, 0xc033, 0xc02f, 0xc034, 0xc02c, 0xc02d, 0xc02e, 0xc02f, 0xc02c, 0xc02d).

3. Signature Algorithms

The image shows a Wireshark packet capture of a TLS handshake. The 'Packet Details' pane on the right highlights the 'Signature Algorithms' field of a 'Client Hello' packet (packet 100). The list of signature algorithms includes:

- Signature Hash Algorithms Length: 22
- Signature Hash Algorithms (11 algorithms)
- Extension: psk_key_exchange_modes (len=2)
- Extension: record_size_limit (len=2)
- Extension: compress_certificate (len=7)
- Extension: encrypted_client_hello (len=88)
- Extension: pre_shared_key (len=28)

The 'Packet Bytes' pane on the right shows the raw data of the signature algorithms field, which is a list of 11 signature algorithm identifiers (0xc02b, 0xc030, 0xc032, 0xc031, 0xc033, 0xc02f, 0xc034, 0xc02c, 0xc02d, 0xc02e, 0xc02f).

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

18282 85.61681622 172.25.6.28 172.16.100.252 TLSv1.3 159 Application Data
 18283 85.61678937 172.25.6.28 172.16.100.252 TLSv1.3 156 Application Data
 18286 85.61691075 172.25.6.28 172.16.100.252 TLSv1.3 158 Application Data
 18287 85.61701804 172.25.6.28 172.16.100.252 TLSv1.3 170 Application Data
 18218 85.61712934 172.25.6.28 172.16.100.252 TLSv1.3 158 Application Data
 18218 85.61714085 172.25.6.28 172.16.100.252 TLSv1.3 1963 client Hello (SNI=padlet.net)
 18218 85.61749124 172.25.6.28 172.16.100.252 TLSv1.3 158 Application Data
 18217 85.61753345 172.25.6.28 172.16.100.252 TLSv1.3 164 Application Data
 18218 85.61767019 172.25.6.28 172.16.100.252 TLSv1.3 172 Application Data
 18221 85.61770592 172.25.6.28 172.16.100.252 TLSv1.3 162 Application Data
 18224 85.61783092 172.25.6.28 172.16.100.252 TLSv1.3 169 Application Data
 18227 85.61794216 172.25.6.28 172.16.100.252 TLSv1.3 159 Application Data
 18229 85.61805637 172.25.6.28 172.16.100.252 TLSv1.3 168 Application Data
 18231 85.61815967 172.25.6.28 172.16.100.252 TLSv1.3 167 Application Data
 18232 85.61826745 172.25.6.28 172.16.100.252 TLSv1.3 158 Application Data
 18235 85.61837214 172.25.6.28 172.16.100.252 TLSv1.3 153 Application Data
 18236 85.61846748 172.25.6.28 172.16.100.252 TLSv1.3 157 Application Data
 18237 85.61857812 172.25.6.28 172.16.100.252 TLSv1.3 188 Application Data
 18238 85.61868586 172.25.6.28 172.16.100.252 TLSv1.3 186 Application Data
 18238 85.61879418 172.25.6.28 172.16.100.252 TLSv1.3 157 Application Data
 18240 85.61876793 172.25.6.28 172.16.100.252 TLSv1.3 132 Change Cipher Spec, Application Data
 18247 85.63171427 172.25.6.28 172.16.100.252 TLSv1.3 132 Change Cipher Spec, Application Data
 18248 85.63218191 172.25.6.28 172.16.100.252 TLSv1.3 188 Application Data
 18249 85.63211807 172.25.6.28 172.16.100.252 TLSv1.3 484 Application Data
 18250 85.63217932 172.25.6.28 172.16.100.252 TLSv1.3 2411 Application Data

Frame 18248: 288 bytes on wire (2248 bits), 288 bytes captured (2248 bits) on interface any, id 0
 Linux cooked capture v1
 Internet Protocol Version 4, Src: 172.16.100.252, Dst: 172.25.6.28
 Transmission Control Protocol, Src Port: 3128, Dst Port: 50384, Seq: 40, Ack: 2767, Len: 212
 Hypertext Transfer Protocol
 Transport Layer Security
 TLSv1.3 Change Cipher Spec, Handshake Protocol: Server Hello
 Content Type: Handshake (22)
 Version: TLS 1.2 (0x0303)
 Length: 128
 Handshake Protocol: Server Hello
 Handshake Type: Server Hello (2)
 Length: 124
 Version: TLS 1.2 (0x0303)
 Random: 7e21db0c142eb0866cb7f4da595459eeef1879c93ad5446f196d626
 Session ID Length: 32
 Session ID: 496d0c2abecf1f9d1cc2f3c342f2d89e21a0811e9c9db3d32550b2993c
 Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)
 Compression Method: null (0)
 Extensions Length: 52
 Extension: pre_shared_key (len=2)
 Type: pre_shared_key (41)
 Length: 2
 Pre-Shared Key extension
 Extension: key_share (len=36) 2x5519

Record Layer (TLS record), 133 bytes

- SYN



Packets: 59188 - Displayed: 6061 (10.2%) - Dropped: 0 (0.0%) Profile: Default

The image shows a Kali Linux desktop environment with several applications open. The primary focus is on the Wireshark network protocol analyzer, which is displaying a packet capture on the 'eth0' interface. The packet list on the left shows a series of TCP segments, with the selected packet (No. 18310) being a SYN packet from 172.16.100.252 to 172.25.6.28. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane at the bottom shows the raw data of the selected packet. Other applications visible on the desktop include a terminal, a file manager, and a web browser showing the Neverssl website. The system clock indicates the date is April 8, 2022.

Apr 8 17:46

Not Secure http://neverssl.com

Apr 8 17:36

*any

No.	Time	Source	Destination	Protocol	Length	Info
17986	85.885873836	172.16.100.252	172.25.6.28	TLSv1.3	13457	Application Data, Application Data, Application Data, Application Data, Application Data, Application...
18018	85.546477914	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18019	85.546549746	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18021	85.546594653	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18022	85.546671844	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18024	85.546711445	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18025	85.546799320	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18026	85.546879528	172.16.100.252	172.25.6.28	TLSv1.3	1516	Application Data
18027	85.546894013	172.16.100.252	172.25.6.28	TLSv1.3	1470	Application Data, Application Data, Application Data
18072	85.607405450	172.25.6.28	172.16.100.252	TLSv1.3	1963	Client Hello (SHI=padlet.net)
18075	85.608049652	172.25.6.28	172.16.100.252	TLSv1.3	200	Application Data
18076	85.608089586	172.25.6.28	172.16.100.252	TLSv1.3	183	Application Data
18079	85.608248377	172.25.6.28	172.16.100.252	TLSv1.3	197	Application Data
18081	85.608378399	172.25.6.28	172.16.100.252	TLSv1.3	160	Application Data
18082	85.608488484	172.25.6.28	172.16.100.252	TLSv1.3	160	Application Data
18084	85.608688925	172.25.6.28	172.16.100.252	TLSv1.3	154	Application Data
18087	85.609369638	172.25.6.28	172.16.100.252	TLSv1.3	157	Application Data
18088	85.609623951	172.25.6.28	172.16.100.252	TLSv1.3	162	Application Data
18090	85.609726625	172.25.6.28	172.16.100.252	TLSv1.3	172	Application Data
18095	85.610169479	172.25.6.28	172.16.100.252	TLSv1.3	1963	Client Hello (SHI=padlet.net)
18096	85.610223475	172.25.6.28	172.16.100.252	TLSv1.3	160	Application Data
18098	85.610297964	172.25.6.28	172.16.100.252	TLSv1.3	163	Application Data
18100	85.610350144	172.25.6.28	172.16.100.252	TLSv1.3	160	Application Data
18101	85.610410472	172.25.6.28	172.16.100.252	TLSv1.3	165	Application Data

TLSv1.3 Record Layer: Handshake Protocol: Client Hello
Content Type: Handshake (22)
Version: TLS 1.0 (0x0301)
Length: 2518

Handshake Protocol: Client Hello (1)
Length: 2514
Version: TLS 1.2 (0x0303)
Random: f22f7c5d6d423da0fb61c6cf8dd1bb6b93cfe09ddae21512f5938b02e7a3
Session ID Length: 32
Session ID: 490e62ab0c1ff9d1cc2df3c24f20d88e210a0611ee9cbdb0e3d32c506b2993c
Cipher Suites Length: 34
Compression Methods Length: 1
Compression Methods (1 method)
Extensions Length: 2407
Extension: server_name (len=35) name=incoming.telenetry.mozilla.org
Extension: extended_master_secret (len=1)
Extension: renegotiation_info (len=1)
Extension: supported_groups (len=16)
Extension: ec_point_formats (len=2)
Extension: application_layer_protocol_negotiation (len=14)
Extension: status_request (len=5)
Extension: delegated_credentials (len=10)
Extension: signed_certificate_timestamp (len=0)

0040 e8 18 0b e2 16 03 01 99 d0 01 00 09 d2 03 03 ef
0050 25 f7 c5 86 4c 23 da d8 fd 01 c6 cf 8d d1 b8 b6
0060 83 e9 fe 09 d0 ae 21 51 2f f3 83 8b c2 e7 a3 2a
0070 49 8e 65 2a b0 fc 1f f9 d1 cc 2d f3 c2 4f 20 d8
0080 8e 21 0a 08 11 ee 9c bd 0e 3d 32 c5 50 b2 99 3d
0090 99 22 13 01 33 03 13 02 c8 2b c9 2f cc a9 cc a8
00a0 c0 2c c0 30 c0 0a c0 09 c0 13 c0 14 00 9c 00 9d
00b0 80 2f 00 35 01 00 09 07 00 00 23 00 21 00 00
00c0 1e 09 6e 03 6f 0d 03 0e 07 2e 74 05 6c 05 d0 05
00d0 74 72 79 2e 6d 6f 7a 09 6c 8c 01 2e 6f 72 67 0d
00e0 17 00 00 ff 01 00 01 00 00 0a 00 10 00 0e 11 c0
00f0 00 1d 00 17 00 18 00 10 01 00 01 01 00 00 02
0100 01 00 00 10 00 0e 00 0c 02 68 32 08 68 74 74 70
0110 2f 31 2e 31 00 05 00 05 01 00 00 00 00 22 60
0120 8a 00 00 04 03 05 03 06 03 02 03 00 12 00 00 00
0130 33 05 2f 05 2d 11 ec 04 c9 40 79 ca 37 85 25 6c
0140 e5 84 44 e1 07 c4 44 2a 0f a5 27 a9 a7 5f 29 60
0150 82 0b 53 16 f3 c4 84 8b 64 0f 3a 75 c0 f6 09 34
0160 eb 0b ad 1b e1 2d 38 12 3b ee 17 85 13 e5 9e 3b
0170 c0 09 3f 9b 48 fe 32 8c 3e 54 57 11 35 7c af 0d
0180 05 6d 59 30 ee c9 93 f5 ba 8a 07 09 34 81 a3 9d
0190 c0 09 87 e7 c5 6d 29 16 9a 80 56 69 40 c2 7f 60
01a0 8a 14 5d 03 c2 1f 83 56 5e 2c a6 11 99 9a 8a 1a
01b0 ae 17 a1 3f 11 a8 b3 9d eb 52 c2 a8 50 79 33 17
01c0 85 fe c2 ec 9e 98 d7 08 8d c9 3b 17 57 1b 06 fb
01d0 05 78 c6 a2 05 70 ec c0 af 05 09 bc c5 11 5f 00

Packets: 59188 - Displayed: 2302 (4.0%) - Dropped: 0 (0.0%) Profile: Default